

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

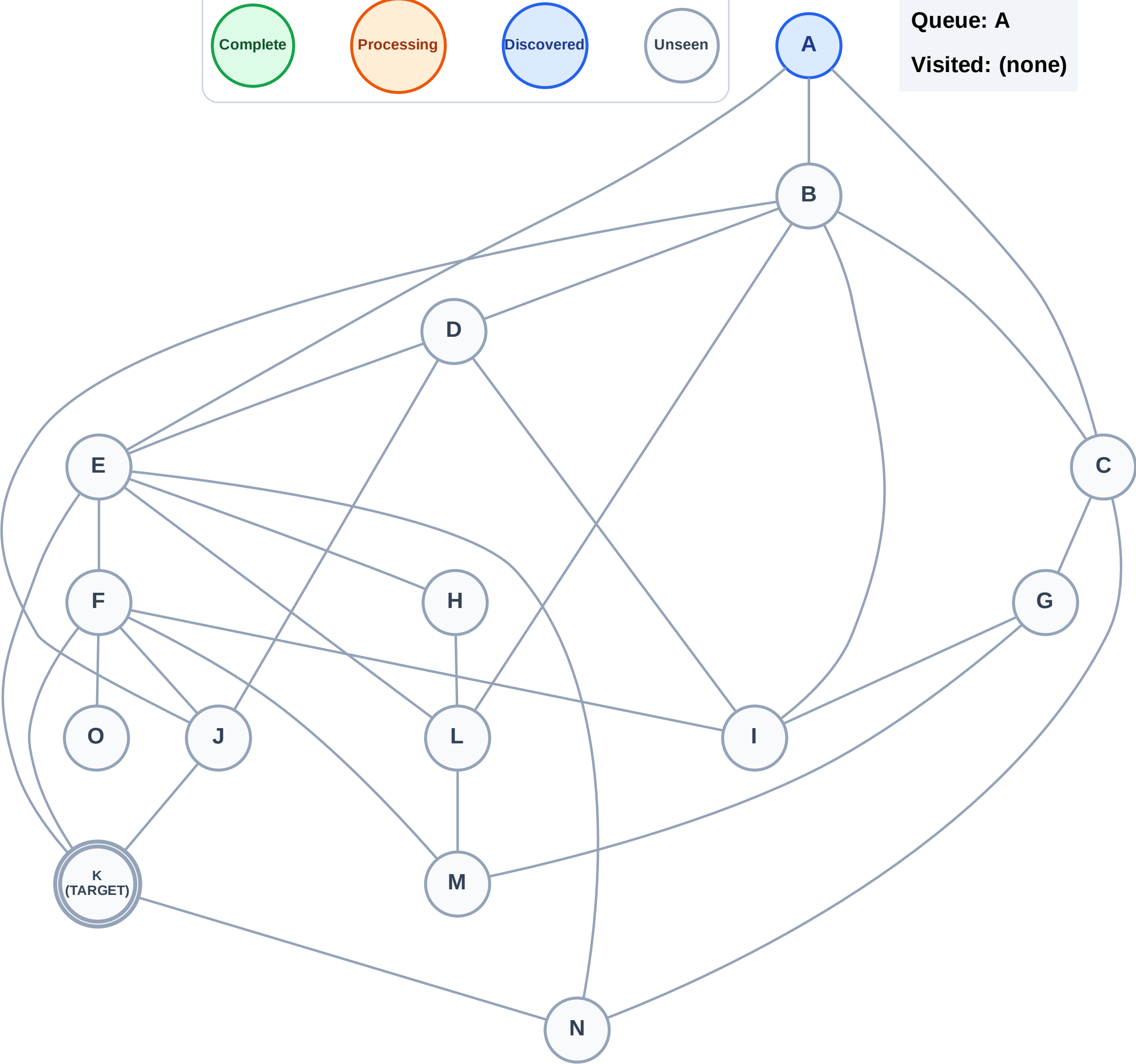
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



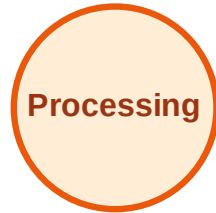
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



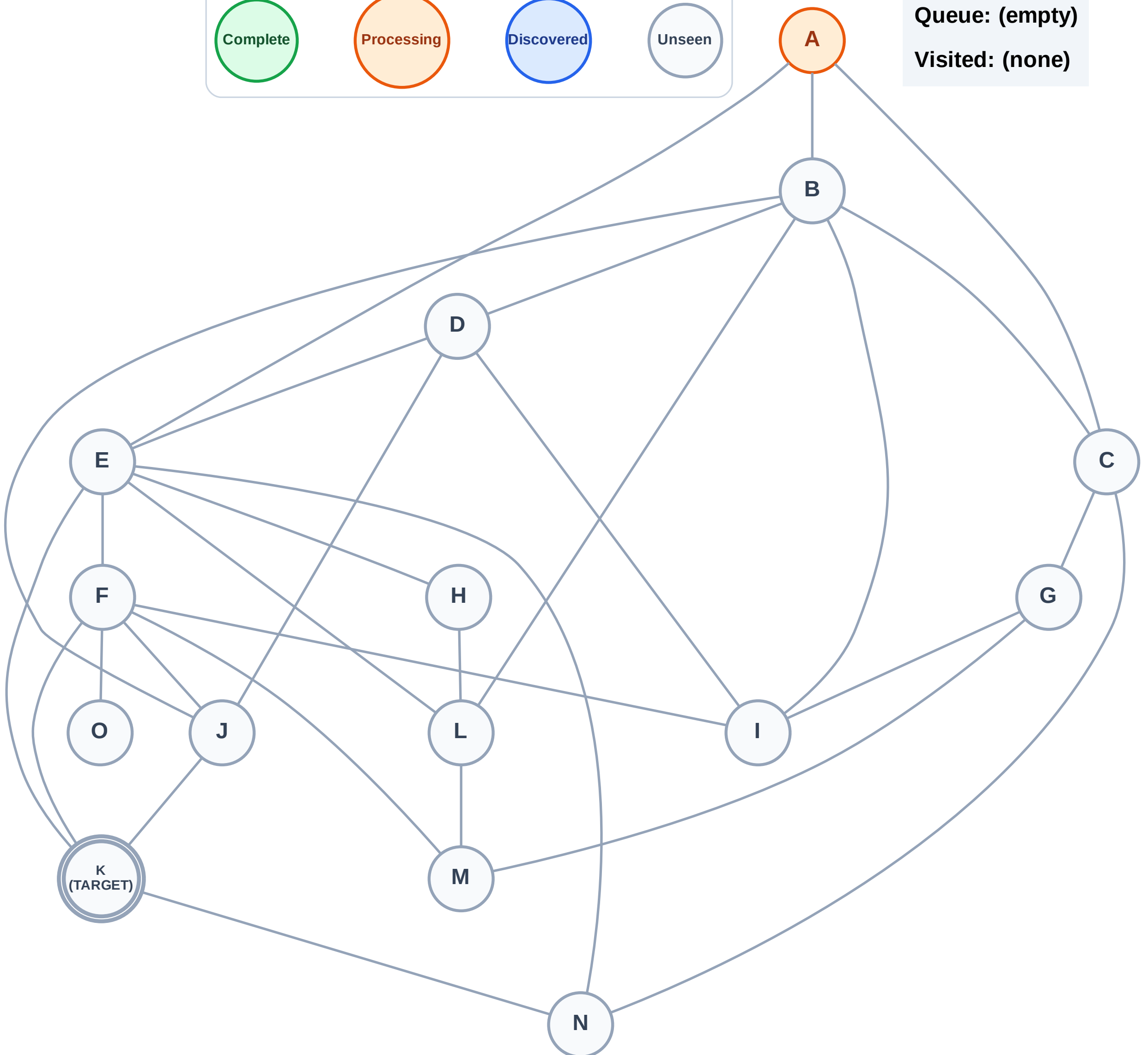
Discovered



Unseen

Queue: (empty)

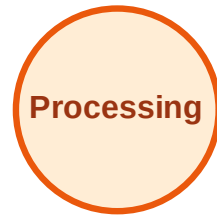
Visited: (none)



BFS Traversal

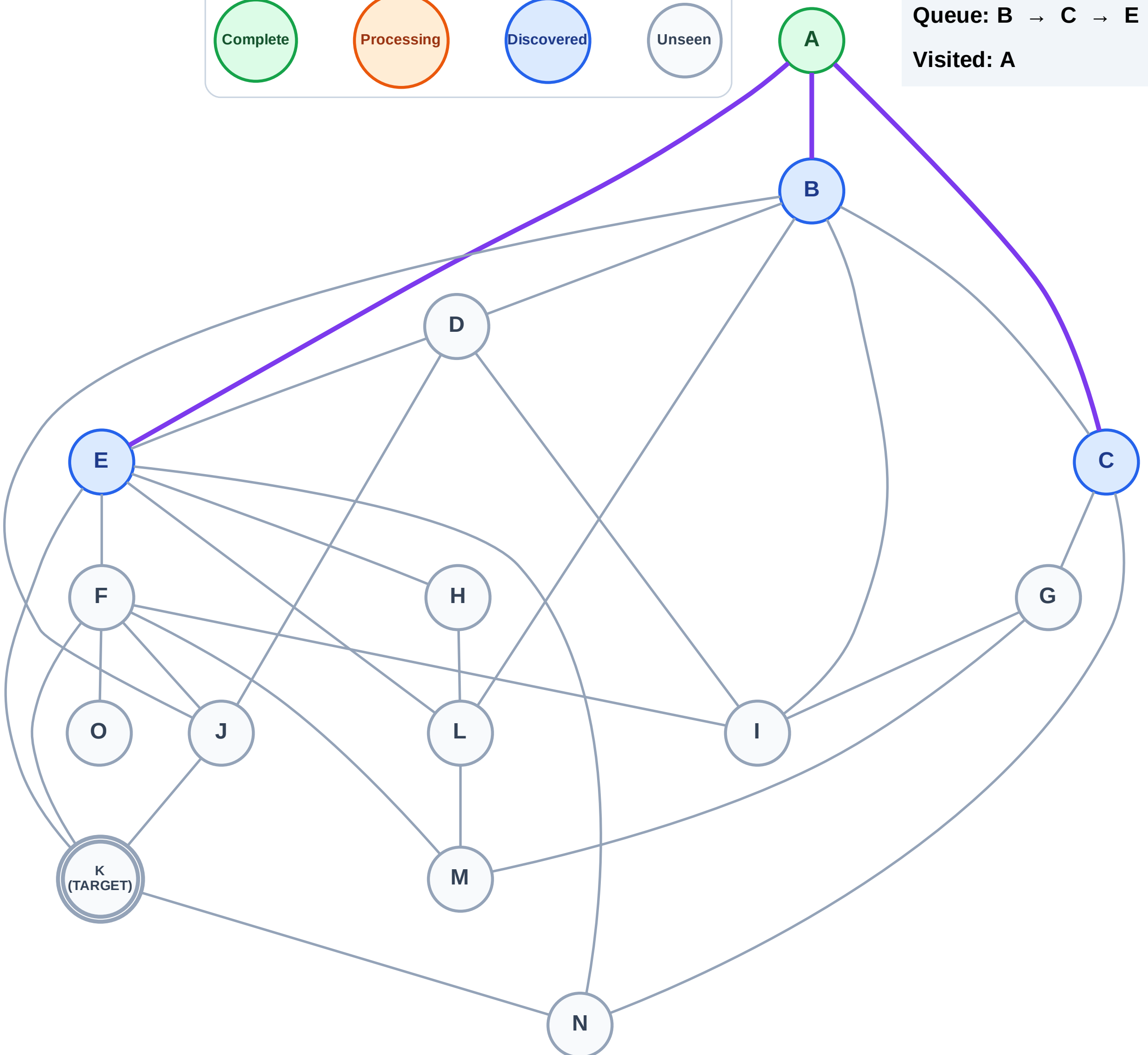
Step 3: Discover B, C, E from A

Legend



Queue: B → C → E

Visited: A



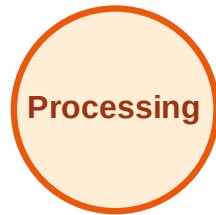
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



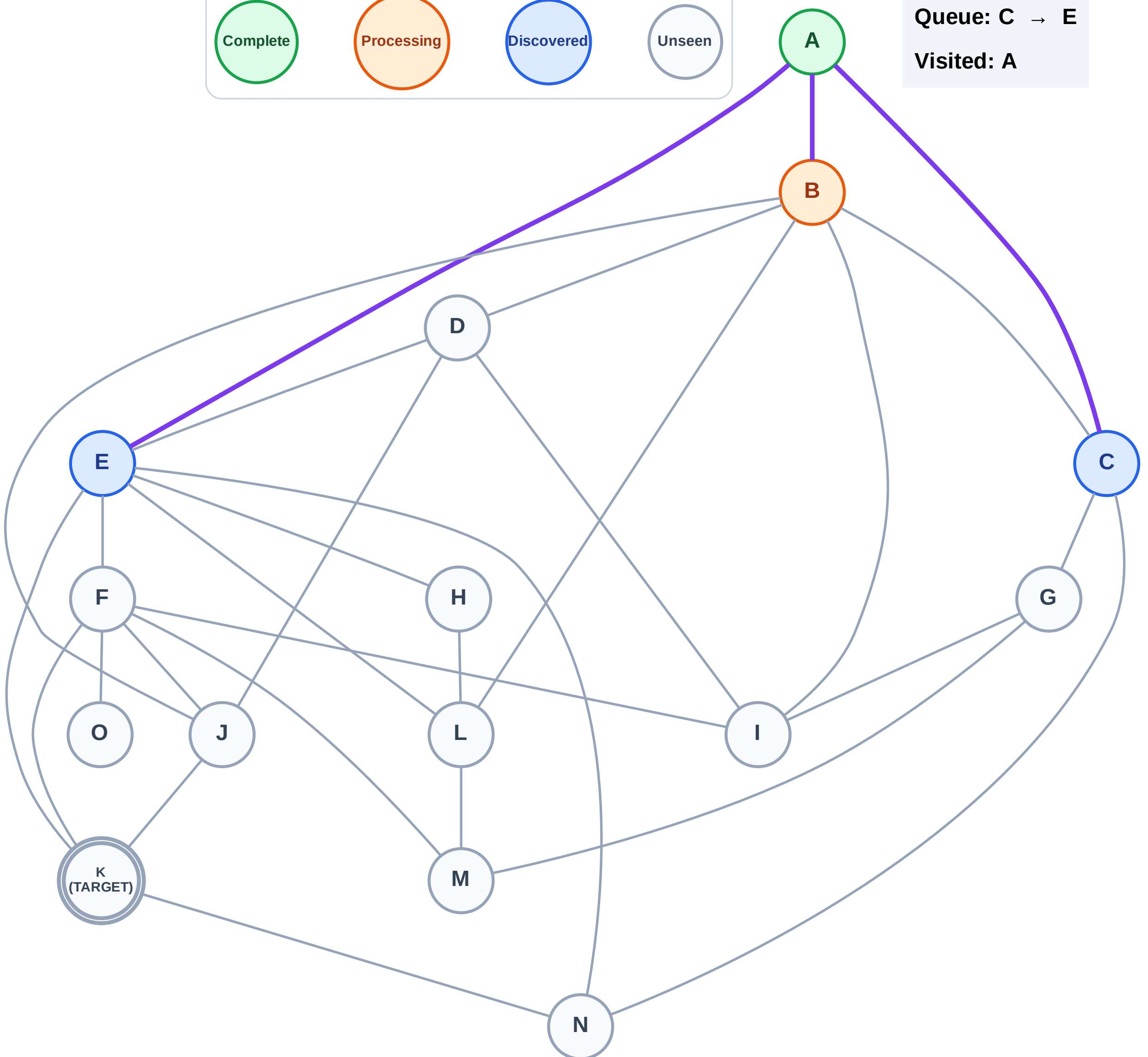
Discovered



Unseen

Queue: C → E

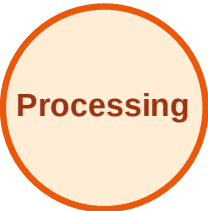
Visited: A



BFS Traversal

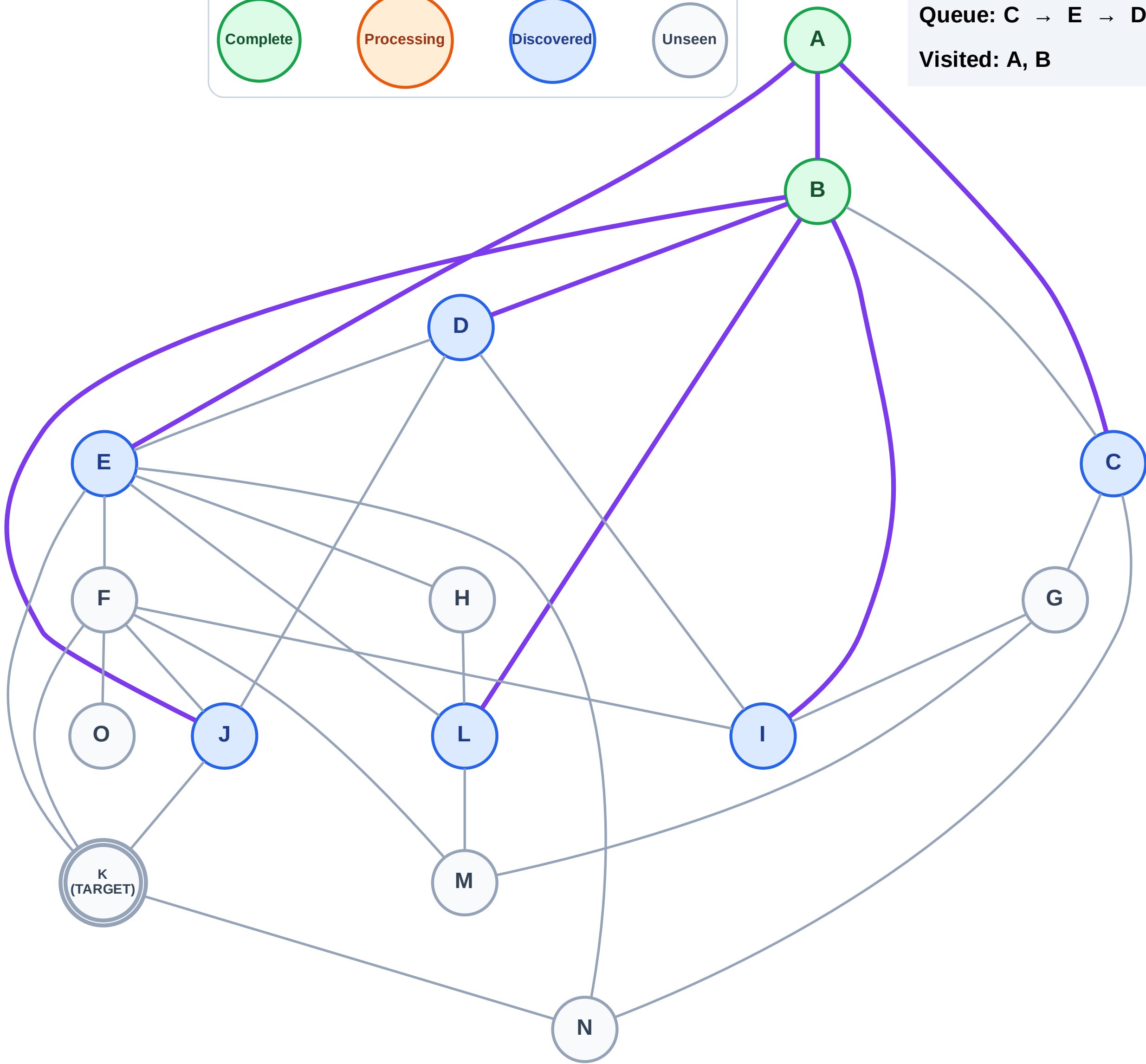
Step 5: Discover D, I, J, L from B

Legend



Queue: C → E → D → I → J → L

Visited: A, B



BFS Traversal

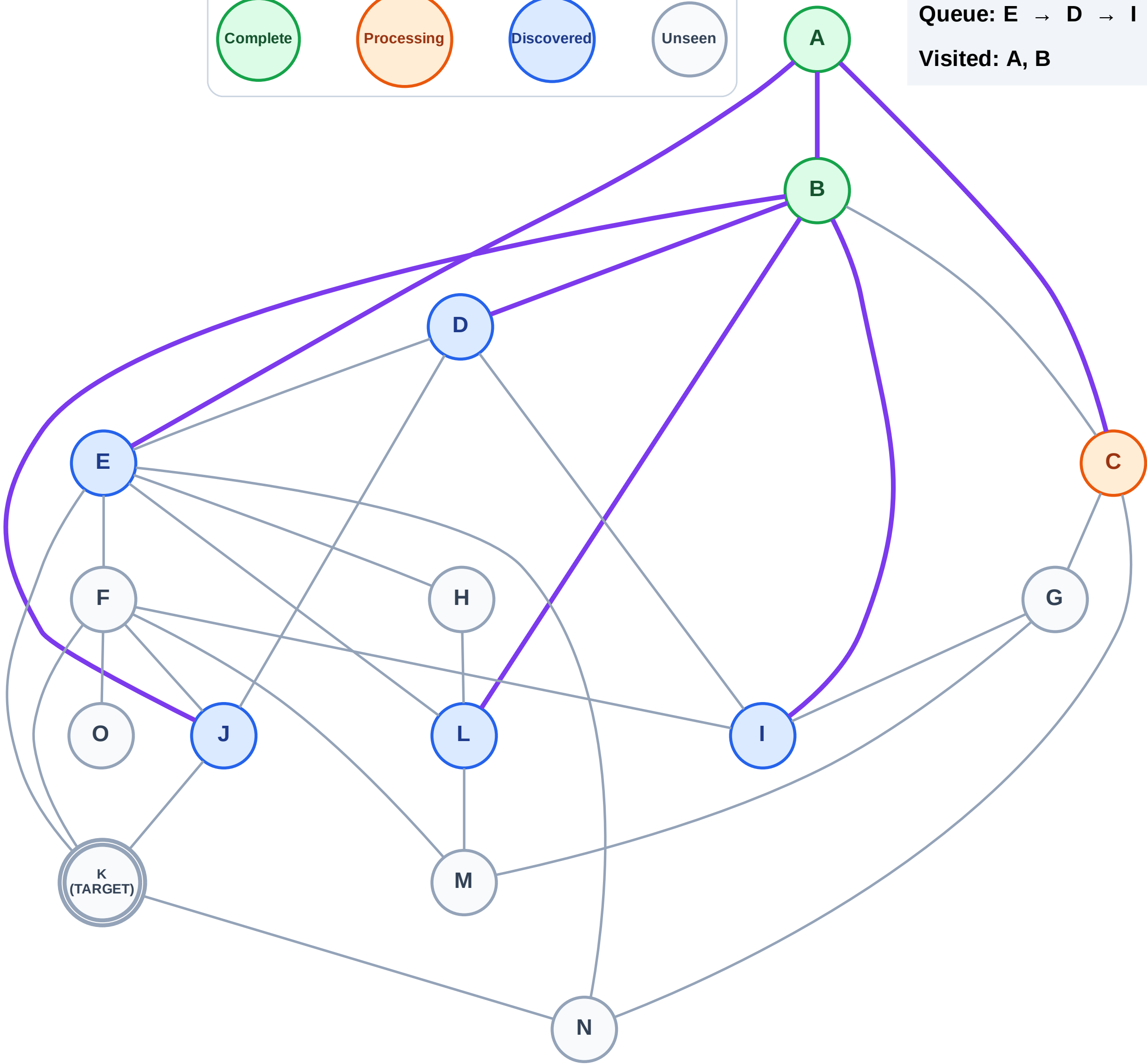
Step 6: Process node C

Legend



Queue: E → D → I → J → L

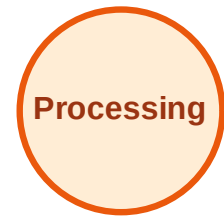
Visited: A, B



BFS Traversal

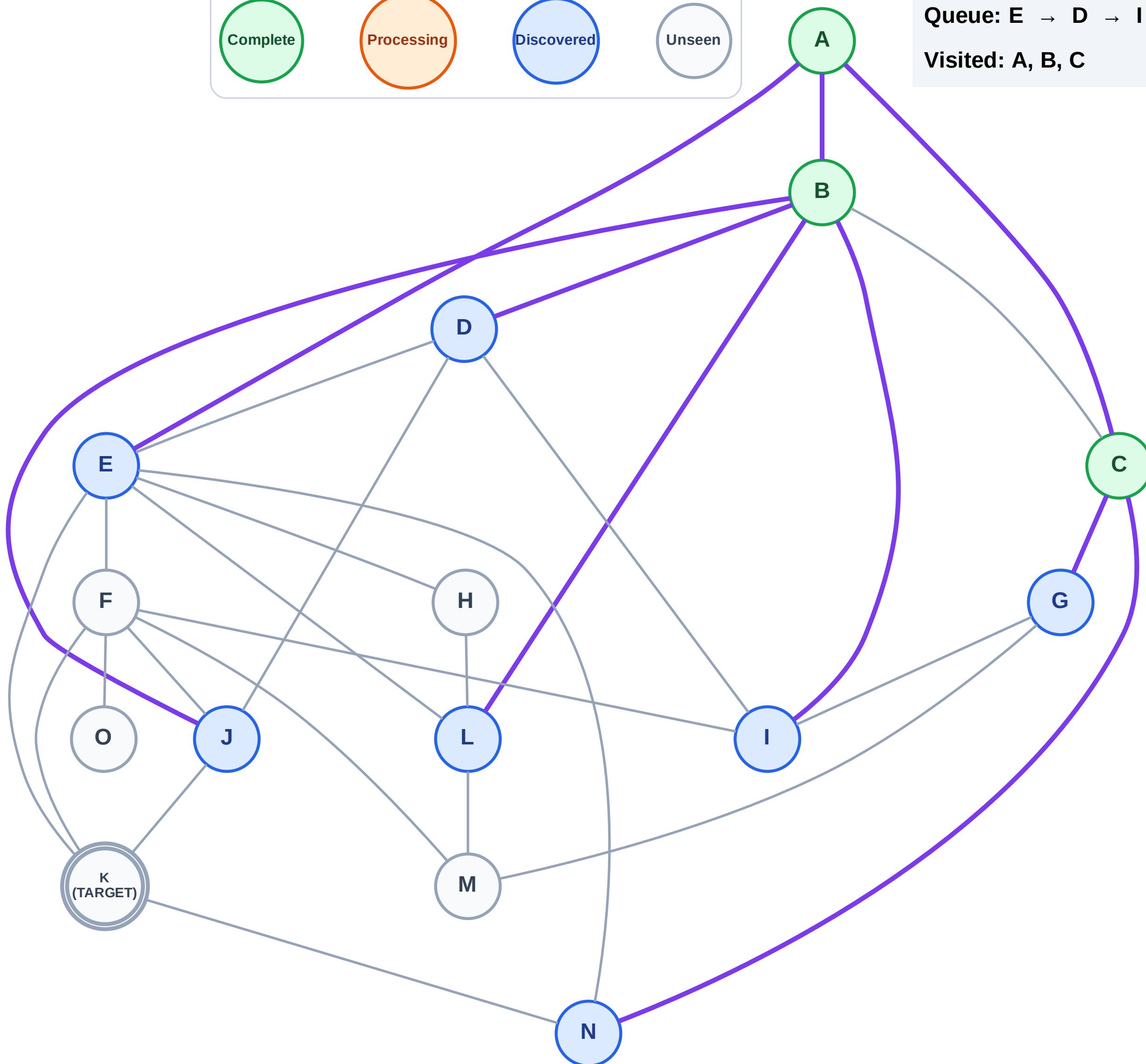
Step 7: Discover G, N from C

Legend



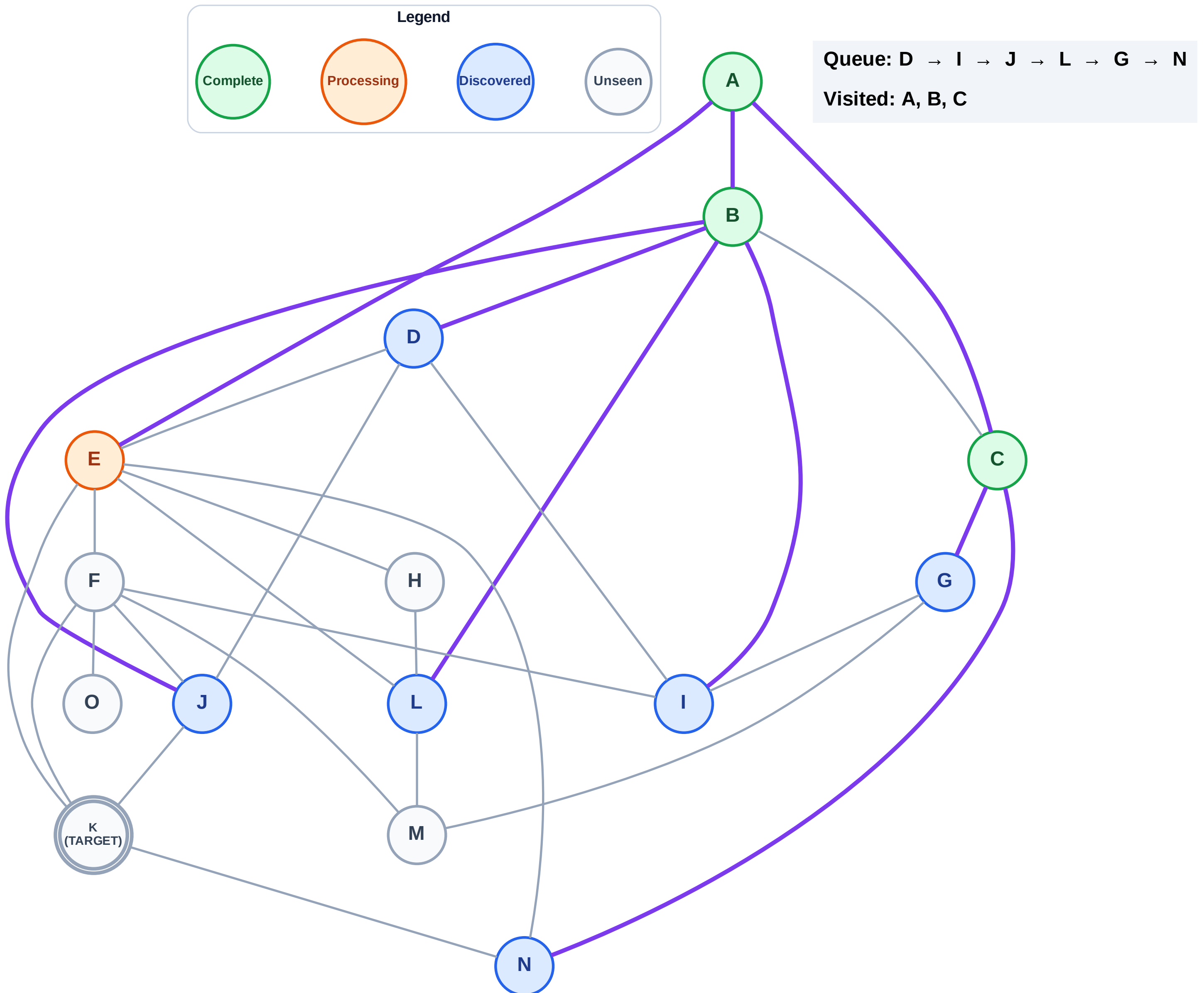
Queue: E → D → I → J → L → G → N

Visited: A, B, C



Step 8: Process node E

Step 8: Process node E



BFS Traversal

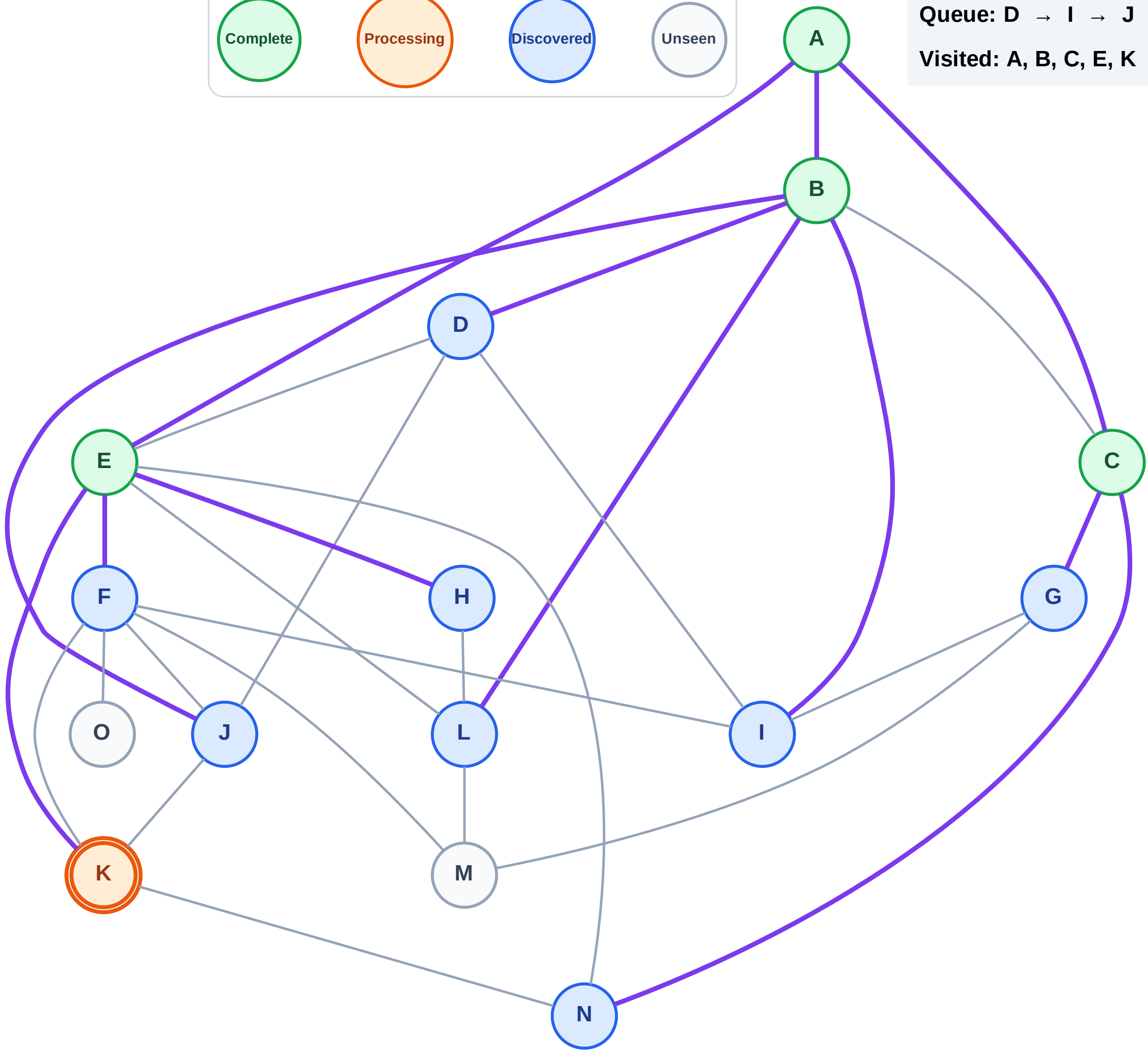
Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend



Queue: D → I → J → L → G → N → F → H

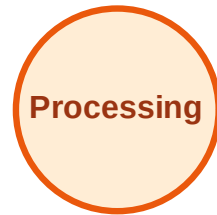
Visited: A, B, C, E, K



DFS Traversal

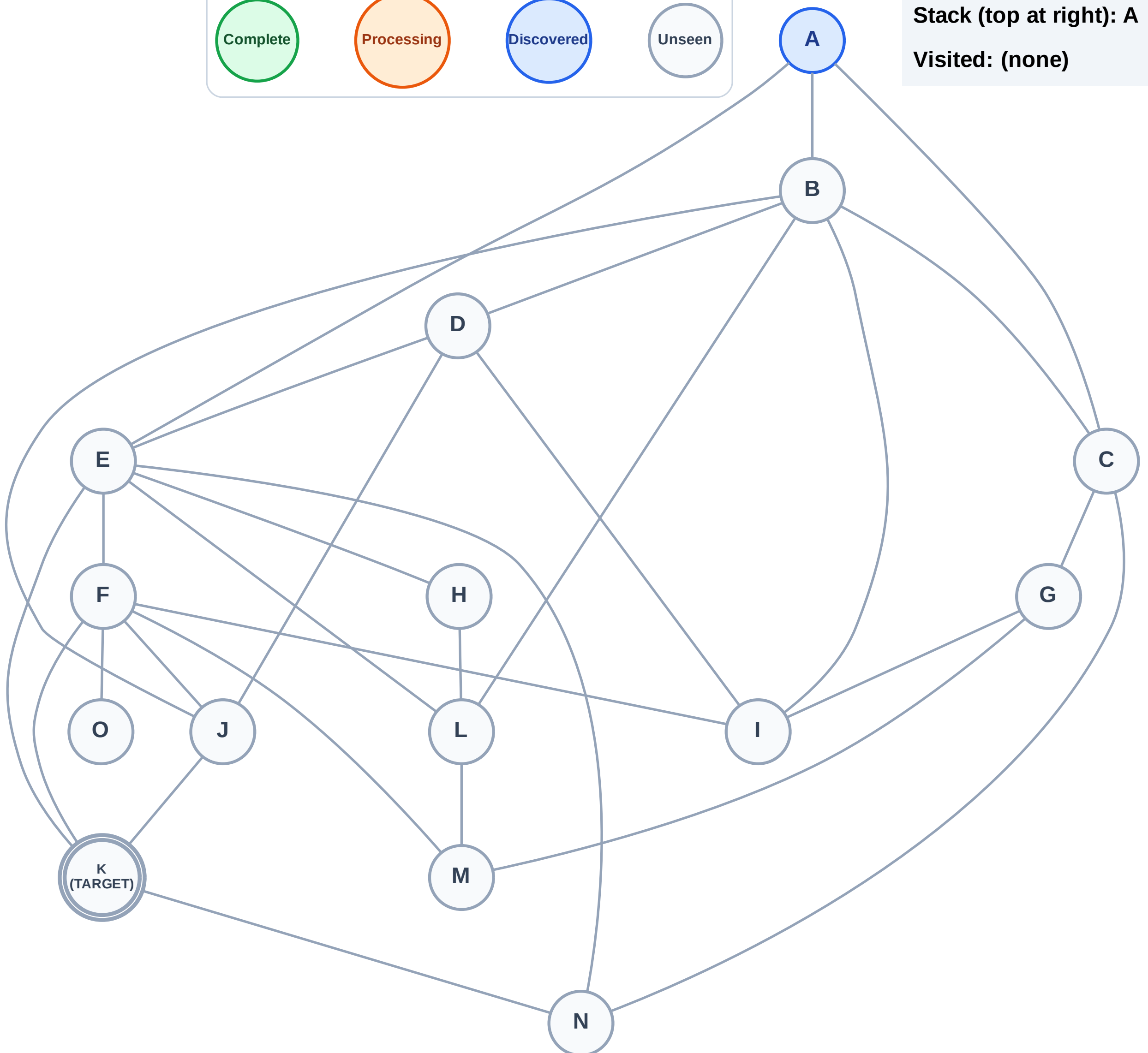
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

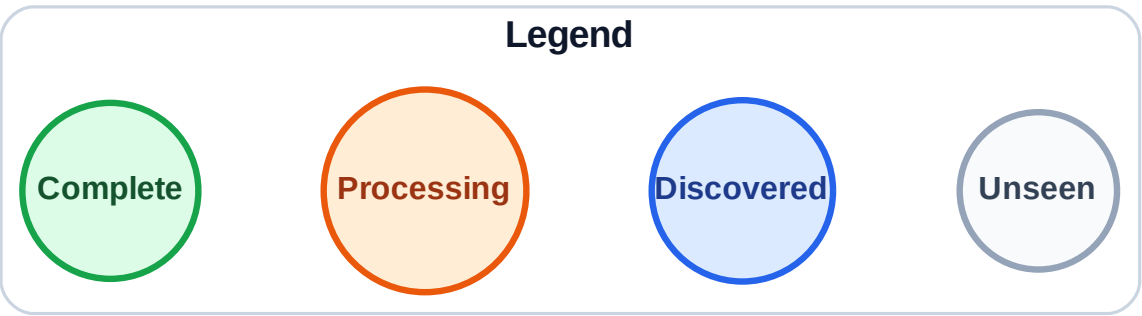
Visited: (none)



DFS Traversal

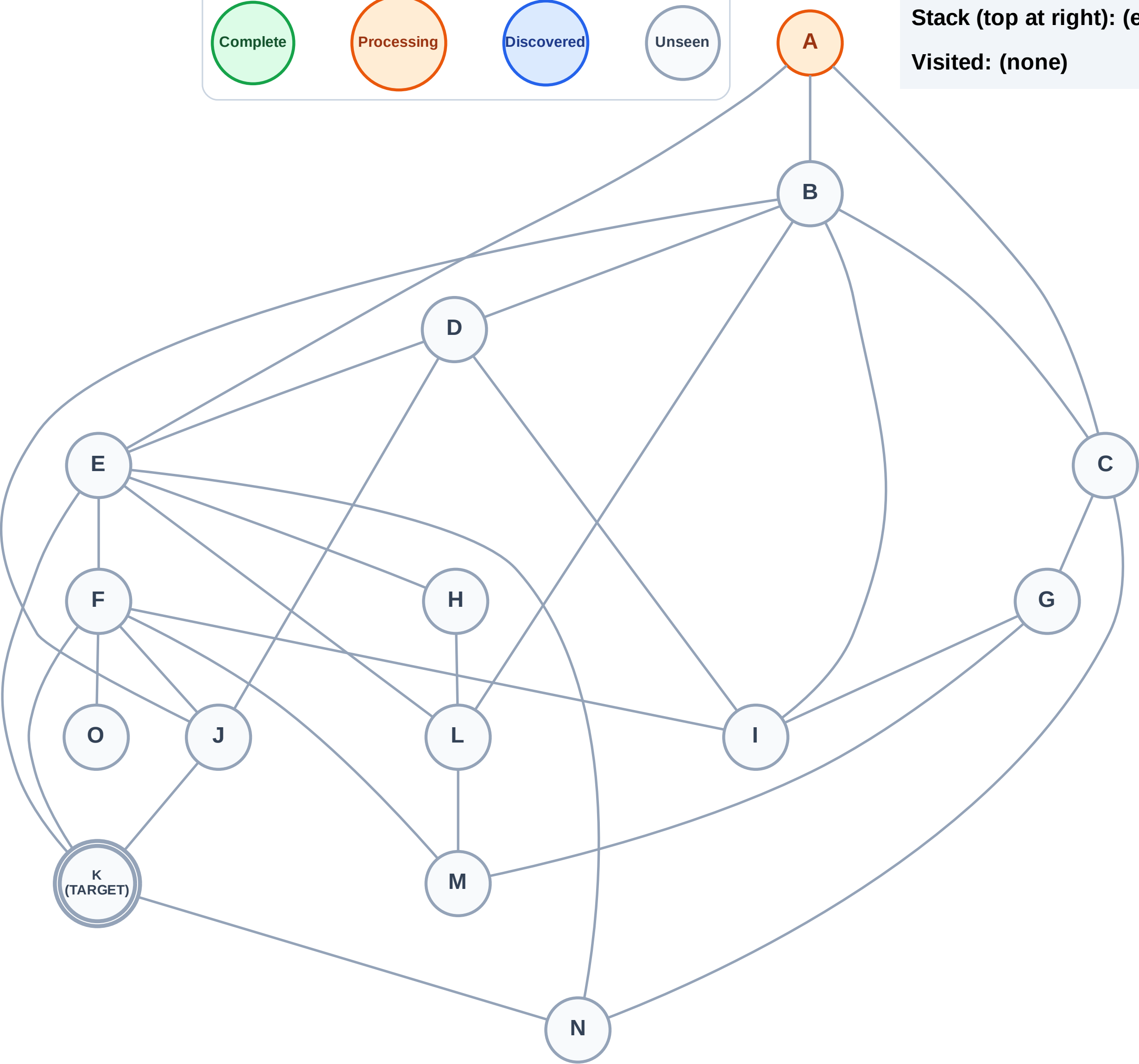
Step 2: Process node A

Legend



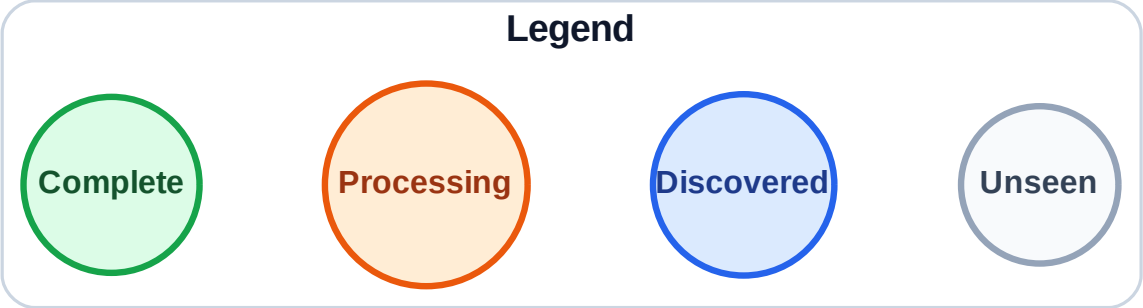
Stack (top at right): (empty)

Visited: (none)



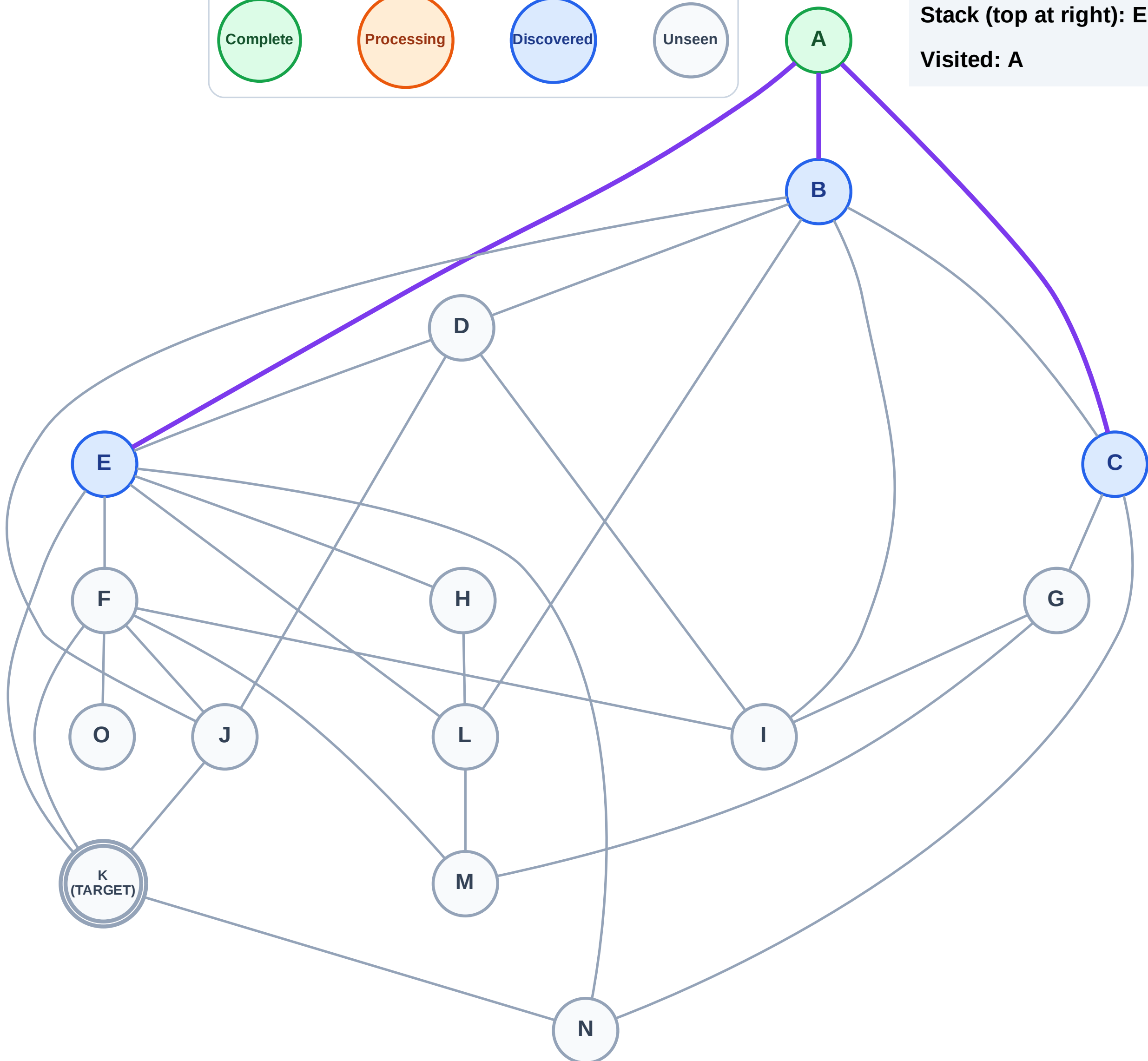
DFS Traversal

Step 3: Discover E, C, B from A



Stack (top at right): E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

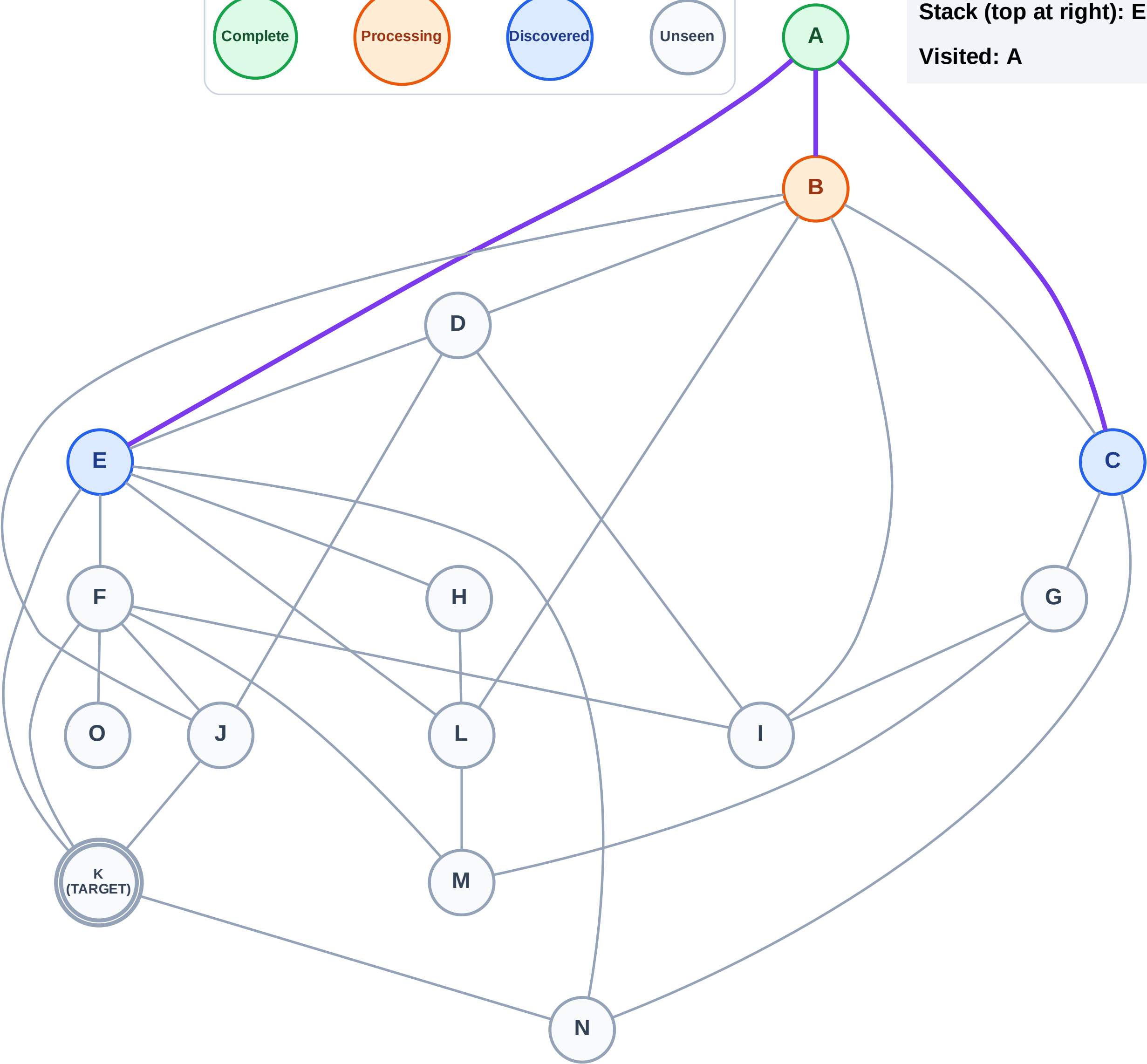
Complete

Processing

Discovered

Unseen

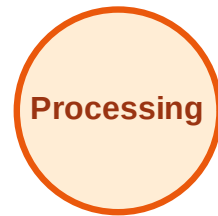
Stack (top at right): E | C
Visited: A



DFS Traversal

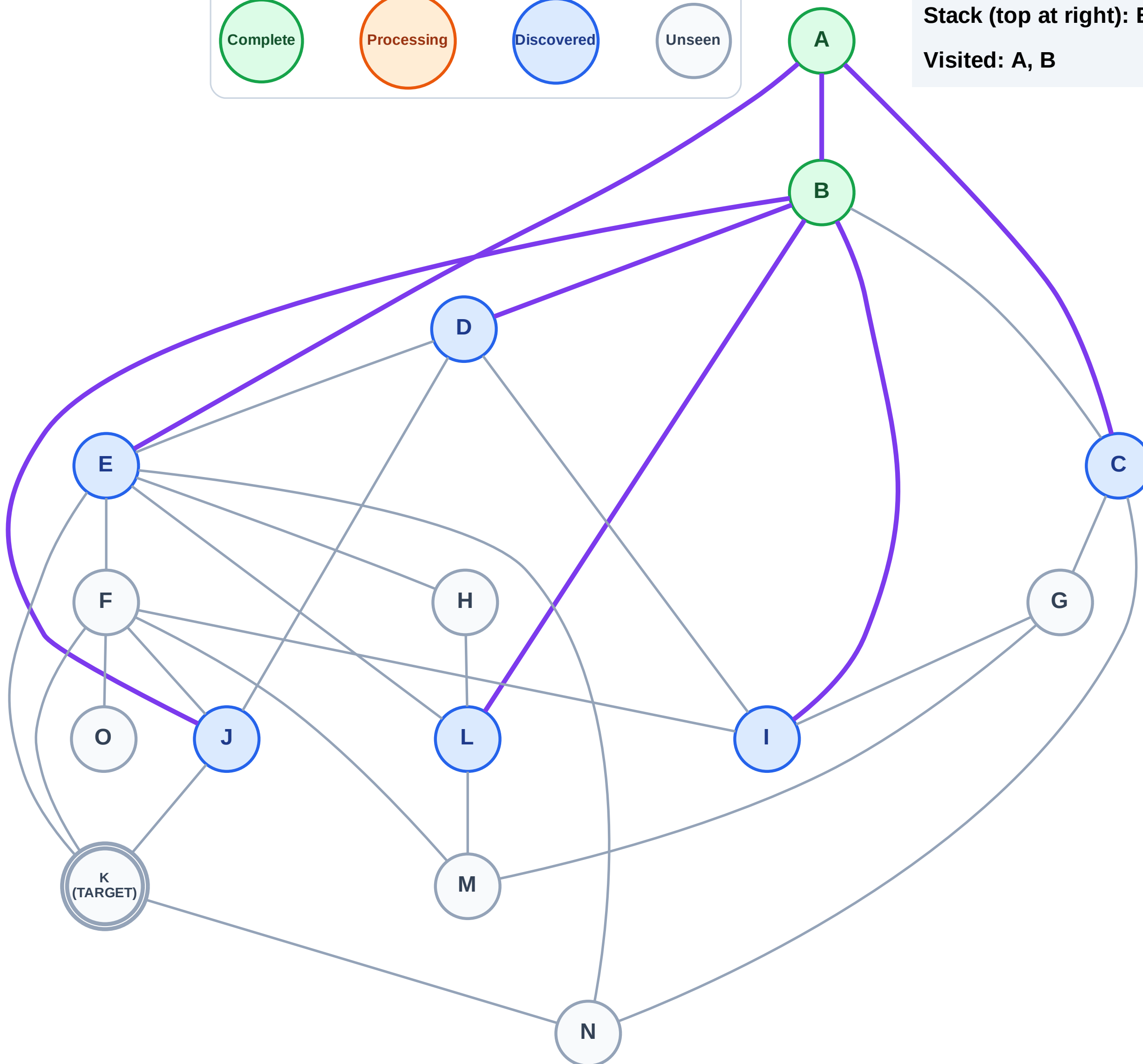
Step 5: Discover L, J, I, D from B

Legend



Stack (top at right): E | C | L | J | I | D

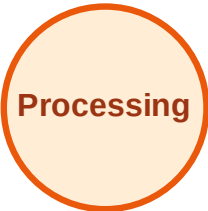
Visited: A, B



DFS Traversal

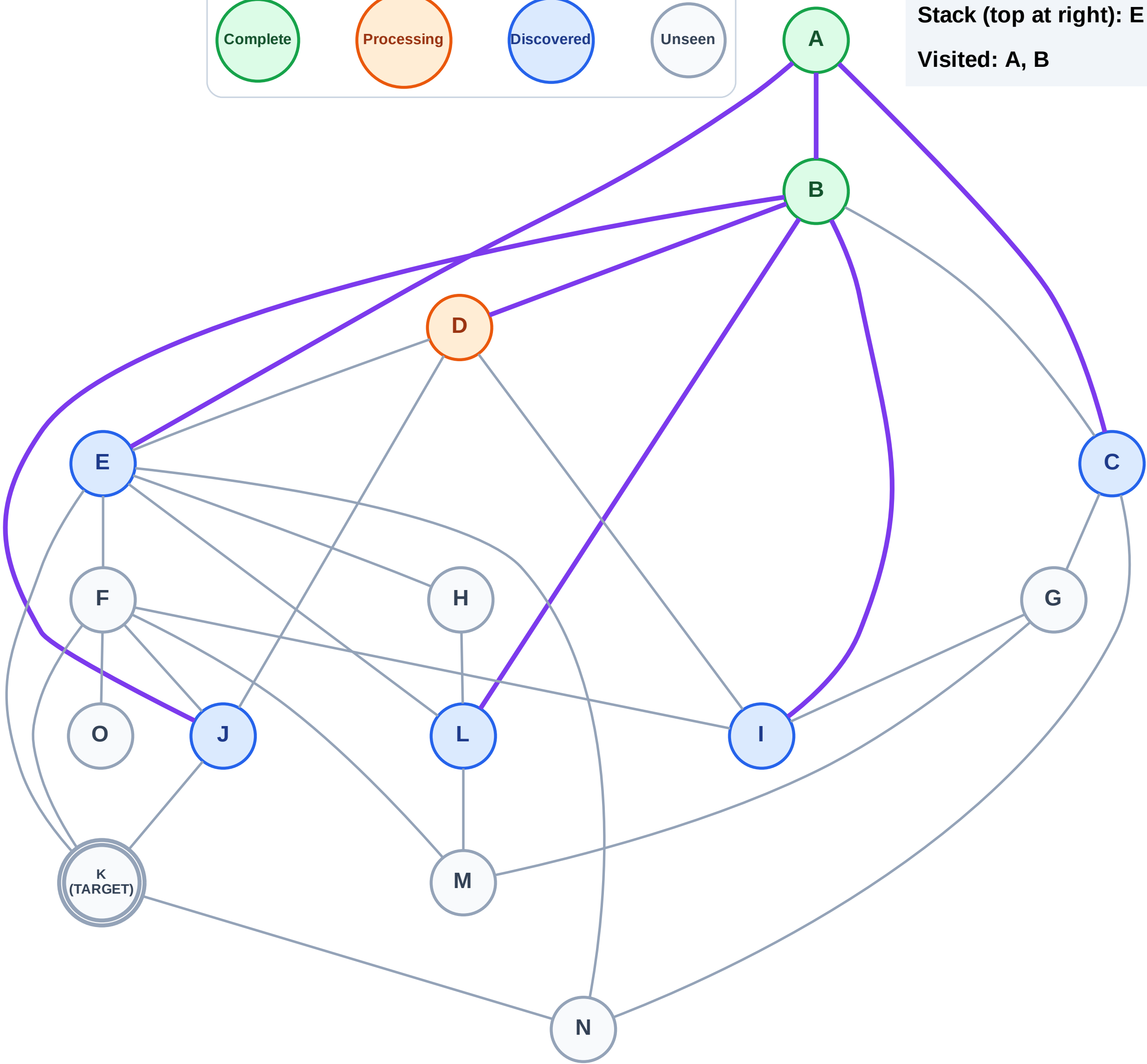
Step 6: Process node D

Legend



Stack (top at right): E | C | L | J | I

Visited: A, B



DFS Traversal

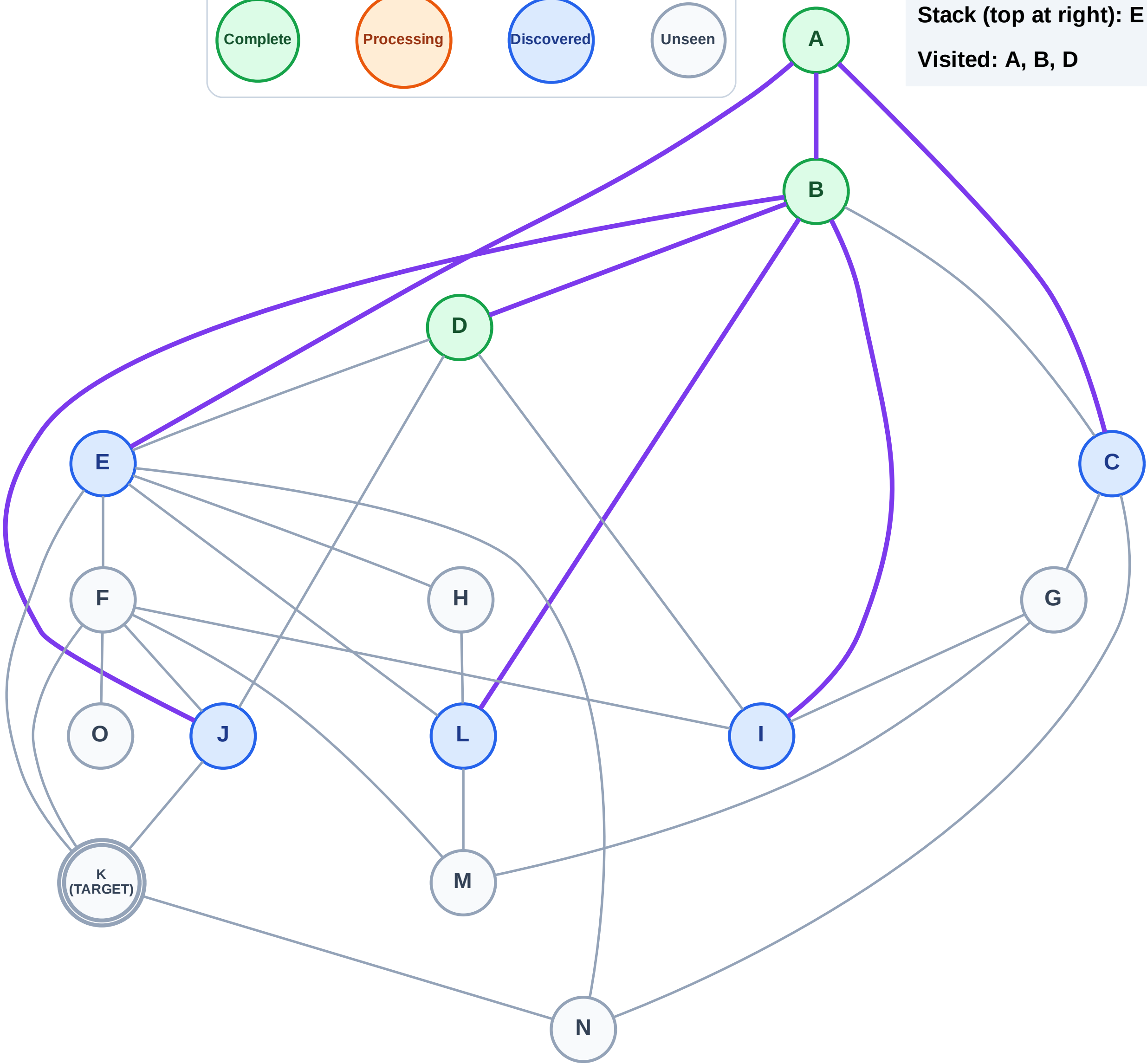
Step 7: No new neighbors from D

Legend



Stack (top at right): E | C | L | J | I

Visited: A, B, D



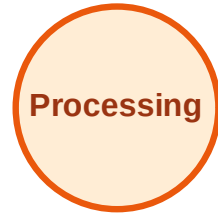
DFS Traversal

Step 8: Process node I

Legend



Complete



Processing



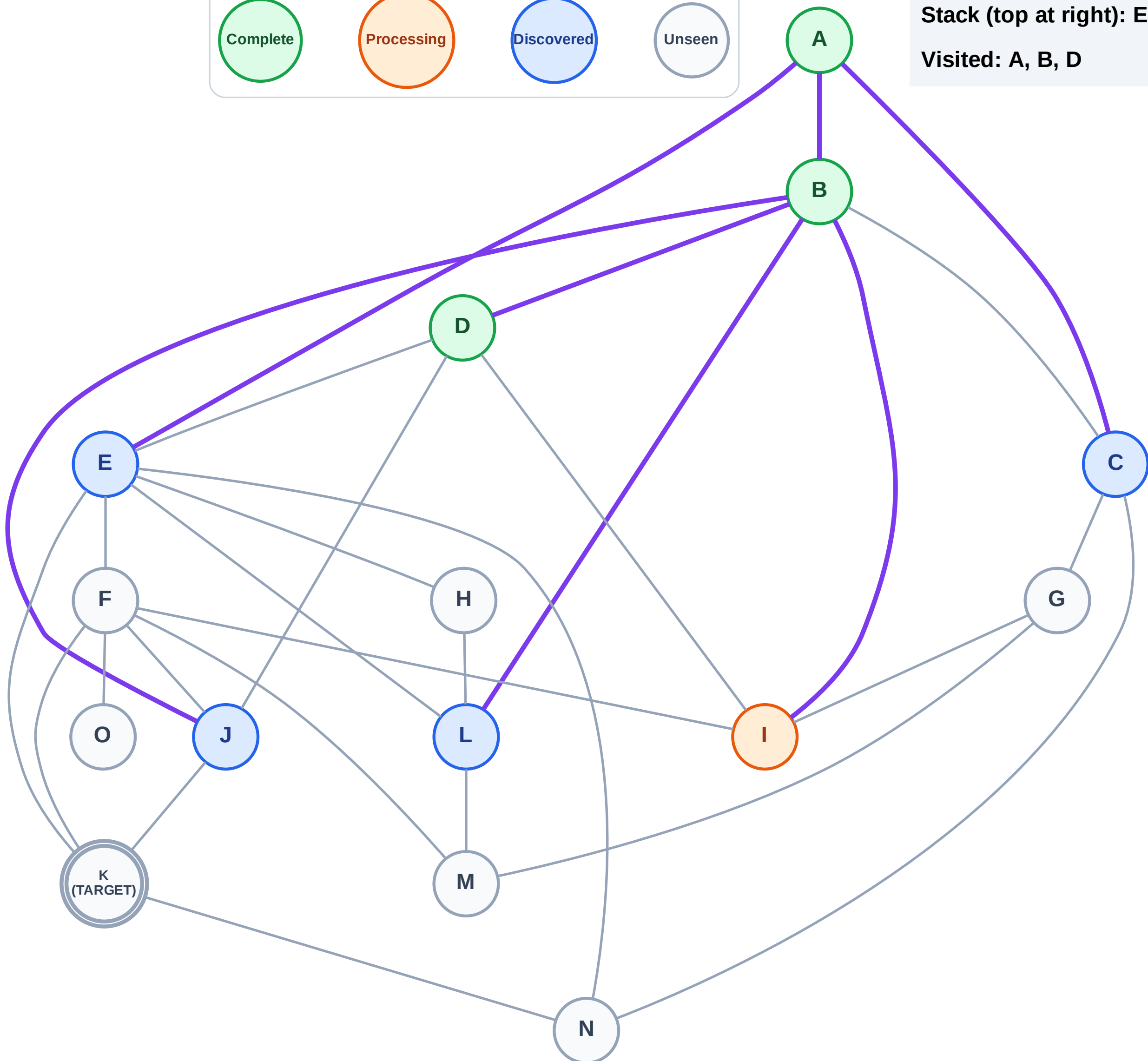
Discovered



Unseen

Stack (top at right): E | C | L | J

Visited: A, B, D



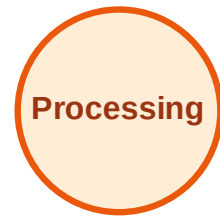
DFS Traversal

Step 9: Discover G, F from I

Legend



Complete



Processing



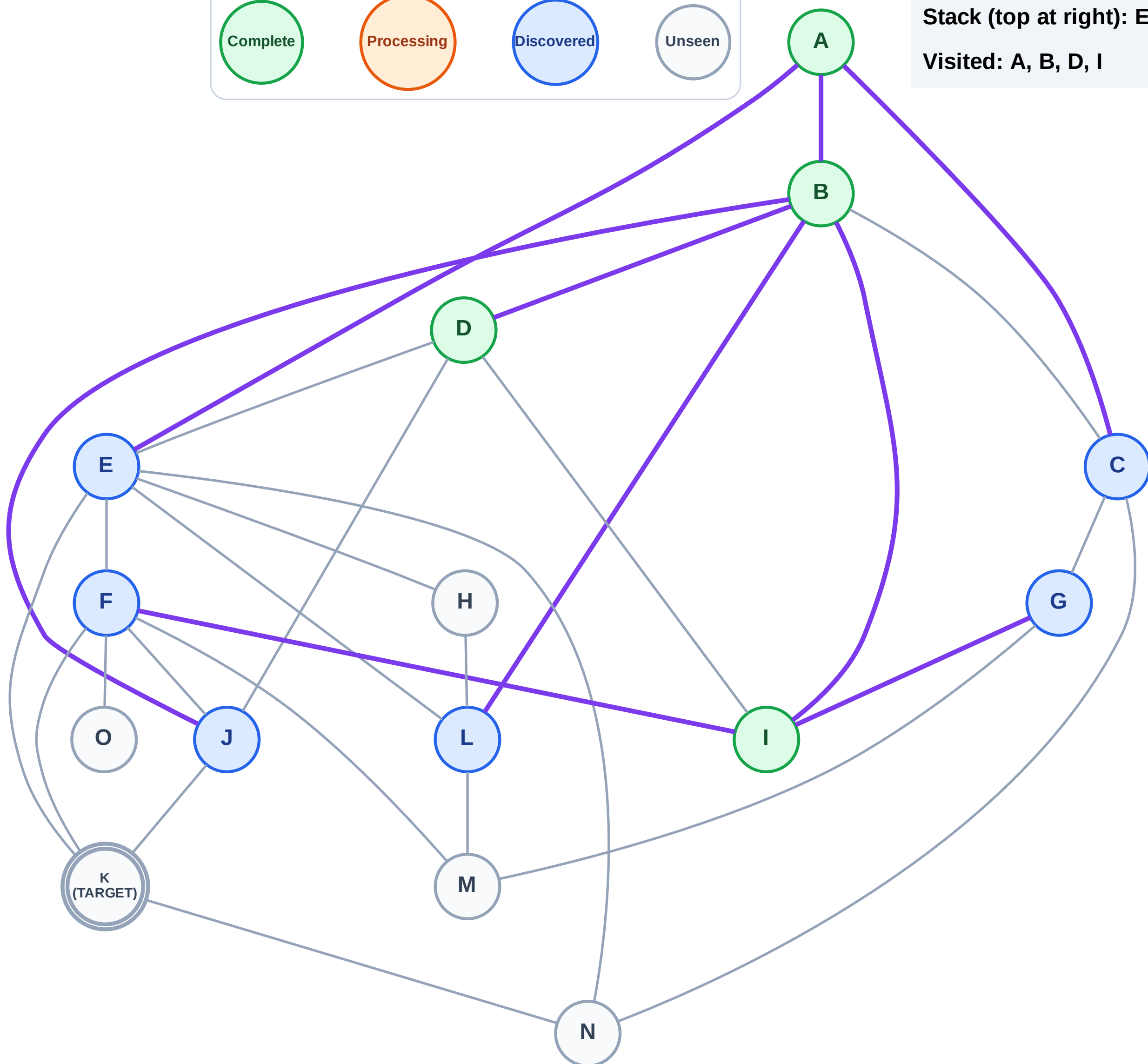
Discovered



Unseen

Stack (top at right): E | C | L | J | G | F

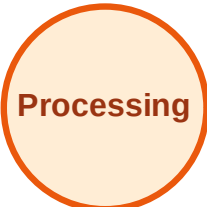
Visited: A, B, D, I



DFS Traversal

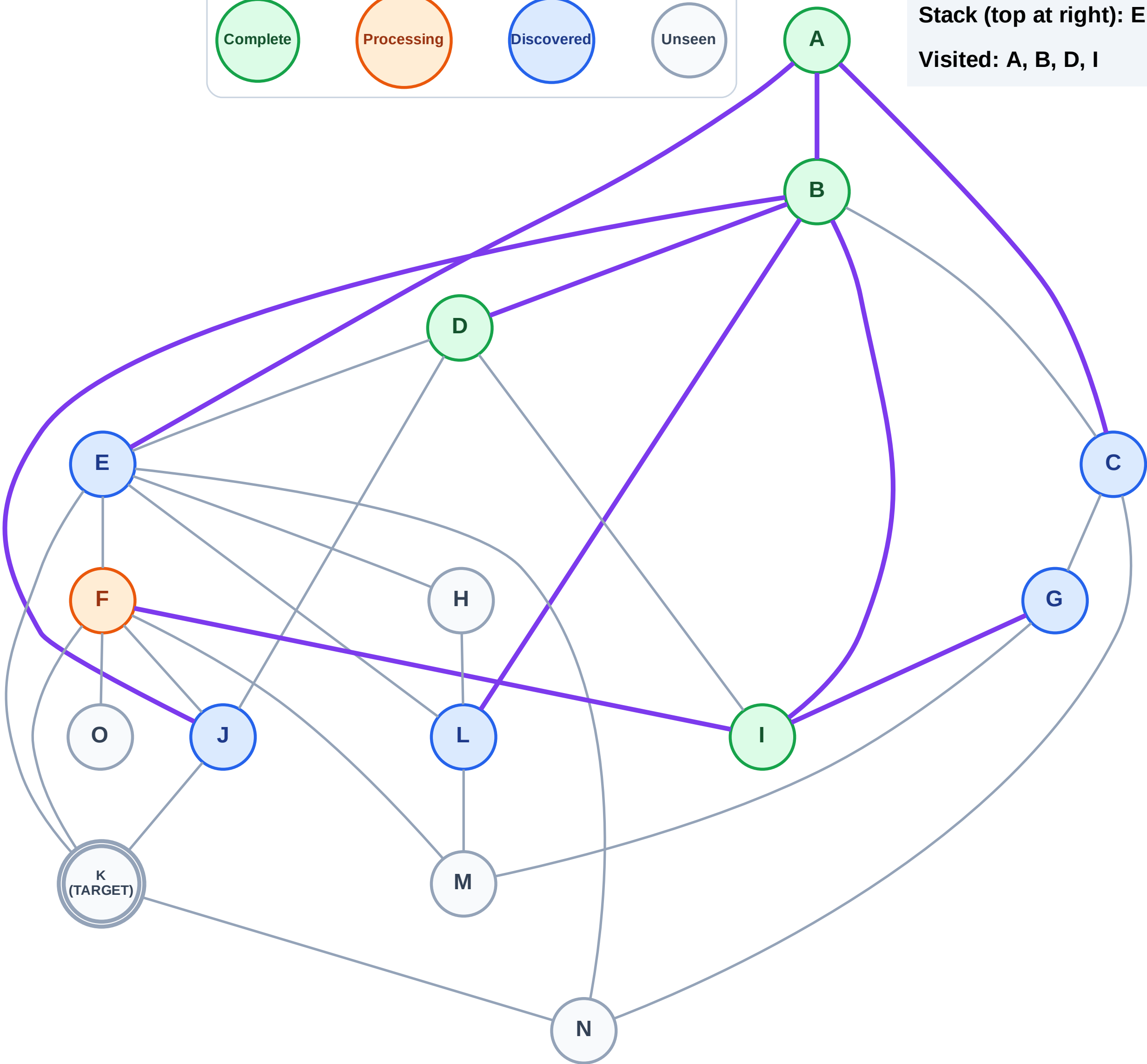
Step 10: Process node F

Legend



Stack (top at right): E | C | L | J | G

Visited: A, B, D, I



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → I → F → K

Legend



Stack (top at right): E | C | L | J | G | O | M

Visited: A, B, D, F, I, K

